

HAT-P-36b

R-1567

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200504 · created 2026-08-01T22:10:51+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458973.812 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1525 +/- 0.0086
Transit depth (Rp/Rs)^2	2.33 +/- 0.26 [%]
Orbital Inclination (inc)	84.19 +/- 1.99
Airmass coefficient 1 (a1)	1.084 +/- 0.016
Airmass coefficient 2 (a2)	-0.062 +/- 0.01
Scatter in the residuals of the lightcurve fit is	1.46 %
Best Comparison Star	#2 - [210, 195]
Optimal Aperture	3.36
Optimal Annulus	12.18
Transit Duration (day)	0.0916 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1525 $\pm$ 0.0086 vs 0.126 (deviation 3.08 σ)
Duration fit vs published	0.0916 d vs 0.0925 d (deviation 0.14 σ)
Mid-time O-C	5.1 min
Depth SNR	17.7
Residual scatter	1.46 %

Light curve

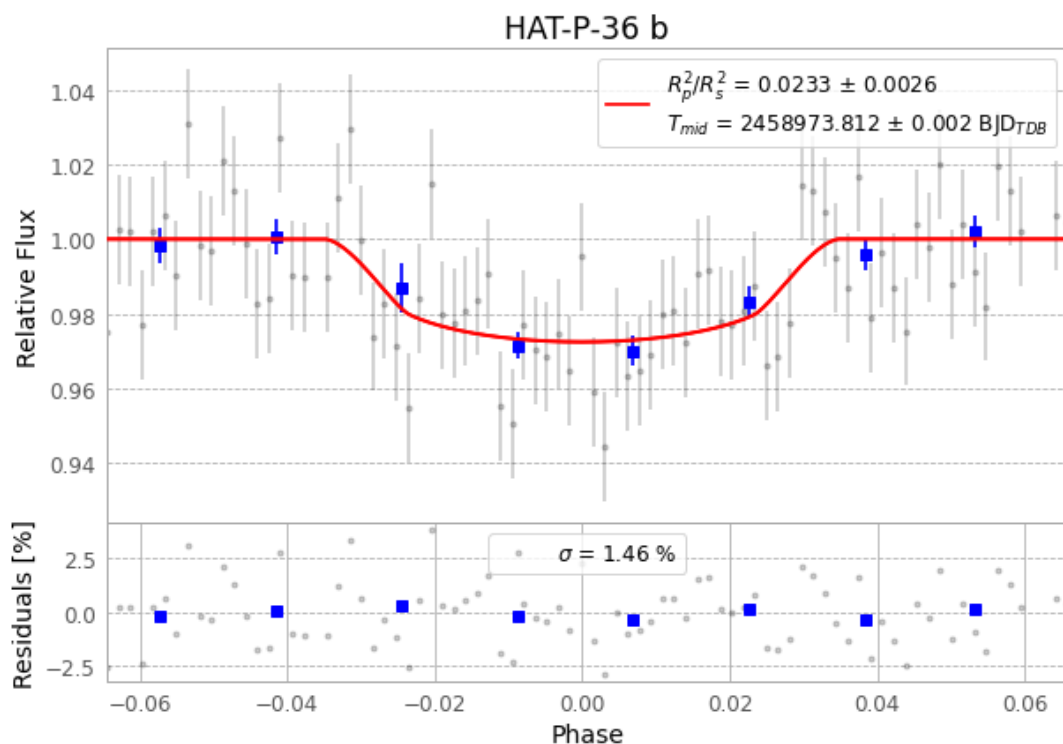


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-05-03.png (SHA-256 aac12195fd3f67ba...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [481, 262], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 878503d73974e89c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

82 FITS frames (54 MB), HATP-36200504052112.FITS → HATP-36200504093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200504.log (27cec1b76e91...), FinalLightCurve_HAT-P-36 b_2020-05-03.png (aac12195fd3f...), FinalLightCurve_HAT-P-36 b_2020-05-03.csv (55a675004aa9...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:10Z.